LUNG IMAGE CLASSIFICATION USING COMPUTER VISION: A DEEP LEARNING APPROACH FOR AUTOMATED DIAGNOSIS

Introduction

Medical imaging plays a critical role in the early detection and diagnosis of lung diseases such as pneumonia and lung opacity. However, the manual interpretation of chest X-ray images is often time-consuming and subject to human error, particularly in resource-constrained healthcare settings where access to experienced radiologists may be limited. With the rapid advancement of Artificial Intelligence (AI), deep learning models especially Convolutional Neural Networks (CNNs) have demonstrated significant potential in automating medical image analysis, improving both efficiency and diagnostic support.

This project presents the development of an AI-assisted image classification system capable of predicting lung conditions from chest X-ray images. The model was trained using Teachable Machine and deployed through Streamlit to create an interactive application that allows users to upload images and receive predictions in real time.

It is important to note that this application is intended strictly for educational and research purposes. It is not a certified medical device and must not be used for clinical diagnosis or decision-making.

Dataset Description
The model was trained using the Kaggle Lung Disease Dataset, which consists of chest X-ray images categorized into three classes:

- Normal: 1,250 images
- Lung Opacity: 1,125 images
- Viral Pneumonia: 1,100 images

To improve model generalization and mitigate overfitting, data augmentation techniques was applied,and this increased the dataset size to 12,278 images:

- Normal: 4,382
- Lung Opacity: 3,969
- Viral Pneumonia: 3,927

These augmentations help simulate real-world variability and improve robustness.

Data Splitting
The dataset was split into:Training set: 70%, Validation set: 15% and Test set: 15%

Methodology
Model Architecture

The model is based on MobileNetV2, a lightweight CNN architecture optimized for efficiency and performance. It is particularly suitable for deployment in real-world applications due to its reduced computational requirements.

Transfer Learning Approach

A transfer learning approach was utilized through the Teachable Machine platform, which leverages pre-trained convolutional neural network architectures (such as MobileNet) for image classification tasks. Rather than training a model from scratch, the system builds upon features learned from large-scale datasets (e.g., ImageNet), enabling efficient learning even with a relatively limited dataset.

All images were: Resized to 224 × 224 pixels and Normalized from pixel range [0, 255] to [-1, 1] This standardization ensured consistent and stable gradient updates during training.

Accuracy and Recall (Sensitivity) Comparison Across All Training Runs

Run	Epoch	Batch size	LR	Viral Pneumonia	Normal	Lung Opacity
1	10	32	0.001	0.99	0.87	0.91
2	15	32	0.001	0.99-0.98	0.80-0.90	0.87-0.87
3	11	32	0.001	0.99-0.99	0.86-0.86	0.92-0.92
4	10	32	0.0008	0.99-0.99	0.90-0.89	0.89-0.92
5	10	16	0.001	0.99-0.99	0.85-0.84	0.91-0.91
6	11	16	0.001	0.99-0.99	0.91-0.91	0.84-0.83
7	**11**	**16**	**0.0009**	**0.99-0.99**	**0.88-0.88**	**0.92-0.92**

Training Strategy

Multiple training experiments were conducted to optimize performance by varying:Epochs, Batch size and Learning rate. The final optimal configuration: Epochs: 11, Batch Size: 16 and Learning Rate: 0.0009

Results and Evaluation

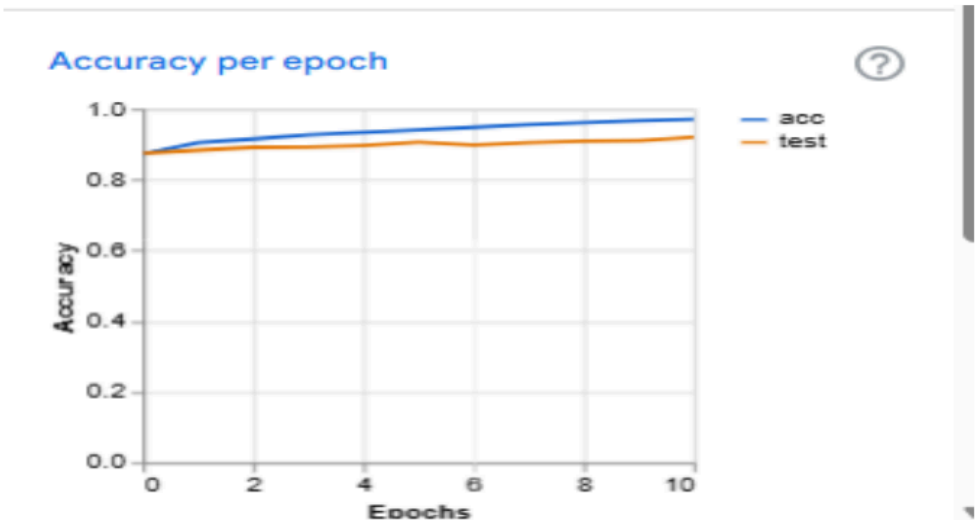
Performance Overview

The model achieved strong classification performance across all classes, with accuracy approaching ~90% on test data.

Confusion Matrix Analysis for LR-0.009 Batch size=16, Epoch(11)

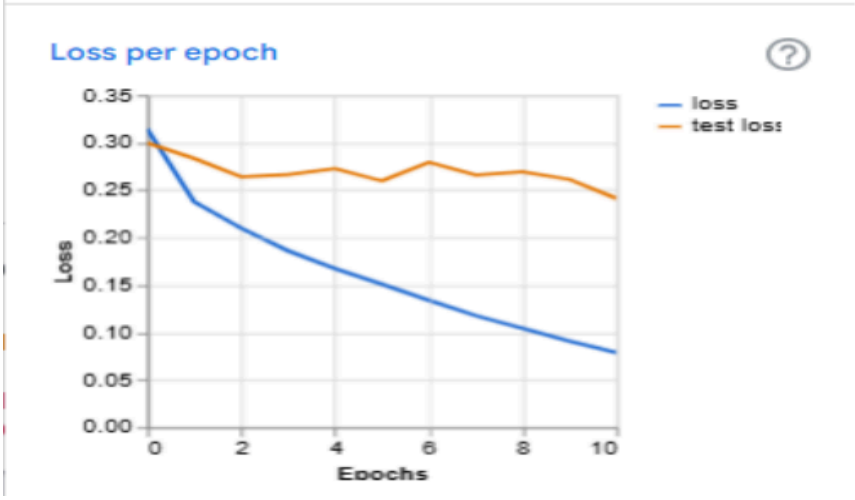
Actual/Predicted	Normal	Lung Opacity	Viral Pneumonia
Normal	573	66	19
Lung Opacity	40	551	5
Viral Pneumonia	5	0	587

11 epochs, batch size 16, learning rate 0.0009



Learning Curve Analysis

1. Training accuracy (blue line): Climbs steadily toward 1.0, showing the model is fitting the training data very well.
2. Test accuracy (orange line): Rises more slowly, leveling around ~0.9 by epoch 10, which indicates good generalization but with a small gap compared to training.



Learning Curve Analysis

1. Training loss (blue line): It steadily decreases across epochs, dropping close to zero. This indicates the model is learning patterns in the training data effectively.
2. Test loss (orange line): It also decreases, though it remains slightly higher than training loss. This gap is expected. It reflects the difference between fitting the training data and generalizing to unseen data

Key Observations

- Viral Pneumonia is classified with very high accuracy
- Some confusion exists between:Normal and Lung Opacity (visually similar features)
- False negatives are minimal, which is important for medical reliability

Interpretation

- The model shows good convergence
- A small gap between training and validation performance indicates mild overfitting
- Overall generalization is acceptable

Discussion

Strengths

- Efficient model (MobileNetV2) suitable for deployment
- Strong performance across all classes
- Effective use of data augmentation
- Good balance between training and generalization

Limitations

- Limited dataset diversity (potential bias)
- Slight confusion between similar classes

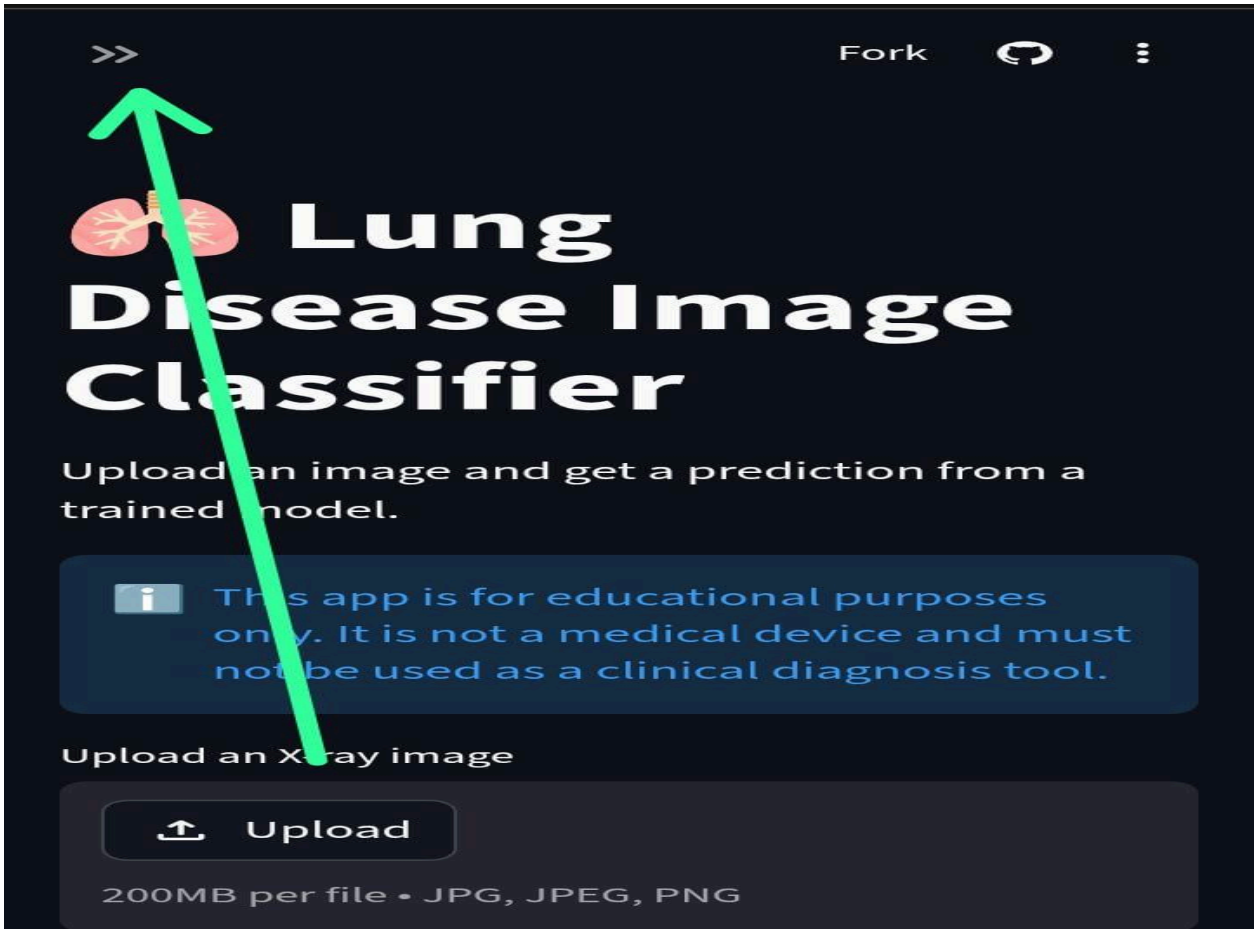
Clinical Implications

- The model can serve as a decision-support tool, not a replacement for clinicians
- High sensitivity is beneficial for early detection
- Misclassification risks must be carefully managed

Deployment

The model was deployed using Streamlit, providing an interactive user interface. You can test here: LUNG DISEASE STREAMLIT APP Also find attached the GIT HUB LINK

Application Features



1. Upload Image Page

- Users upload chest X-ray images
- Model predicts the class
- Displays predicted label with confidence score

2. Sample Images Page

- Preloaded images for demonstration
- Allows testing without external data

User Workflow

1. Upload or select an image
2. Model processes input
3. Prediction is displayed instantly

Conclusion

This study demonstrates the effectiveness of deep learning, specifically MobileNetV2, in classifying lung diseases from chest X-ray images. The model achieved strong performance with good generalization, highlighting the potential of AI in medical imaging.

While promising, the system requires further validation and enhancement before real-world clinical deployment. This represents an important step toward AI-assisted healthcare solutions.

Team members:

1. Asma'u Muhammad Nasiru and Munirat Zubairu trained the model
2. Aliyu Bashir And Abdulkadir Akanbi Sidiq deployed
3. Abdulkadir Akanbi Sidiq and munirat worked on the technical report
4. Asma'u and Aliyu Bashir reviewed the technical report